CLASSIFICATION

COUNTING THE COST – PART III

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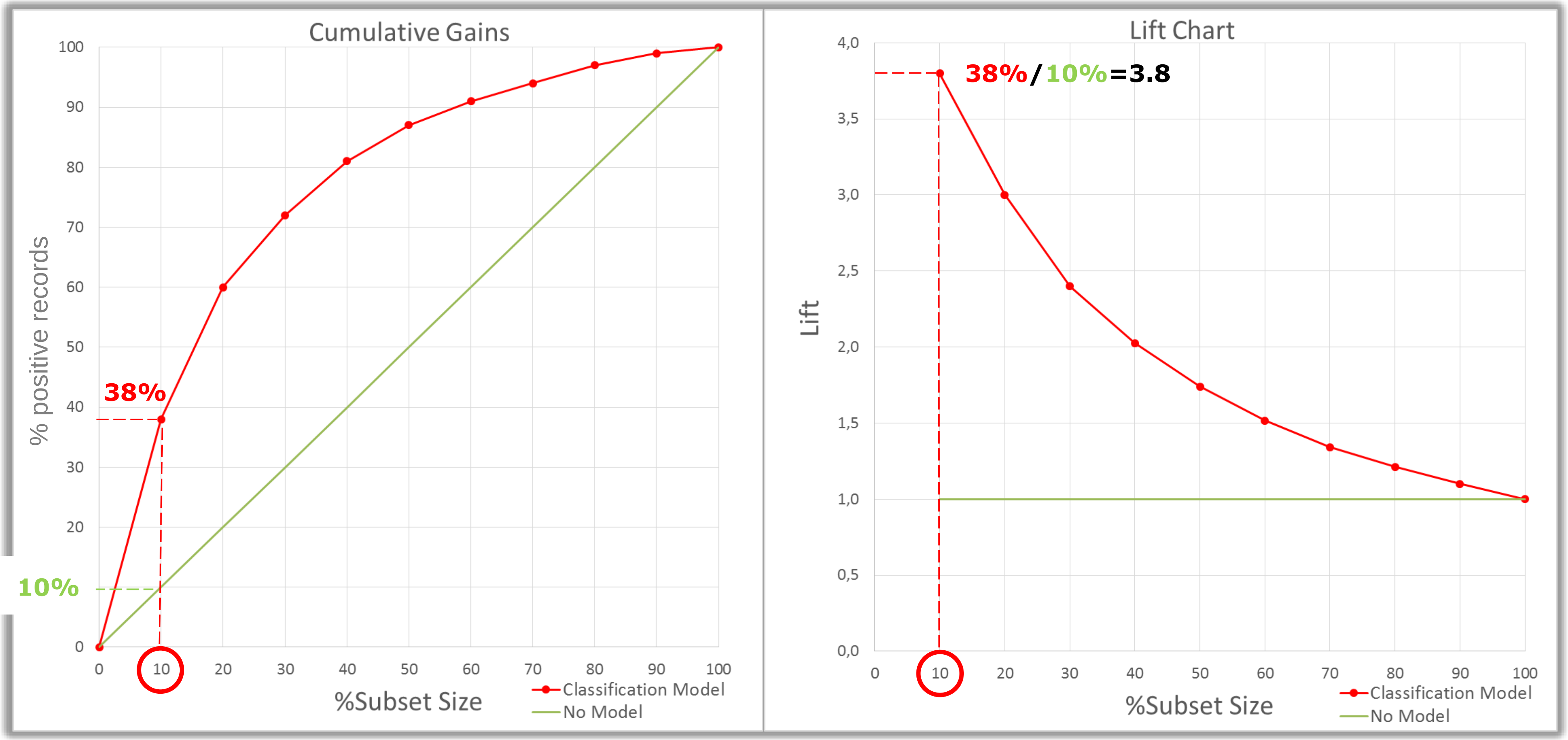
Counting the Cost

The following concepts will be introduced:

- ✓ **Lift Charts**
- ✓ **ROC Curve**

COUNTING THE COST: LIFT CHART

The **CUMULATIVE GAINS** can be mapped to the **LIFT CHART**.



COUNTING THE COST: LIFT CHART

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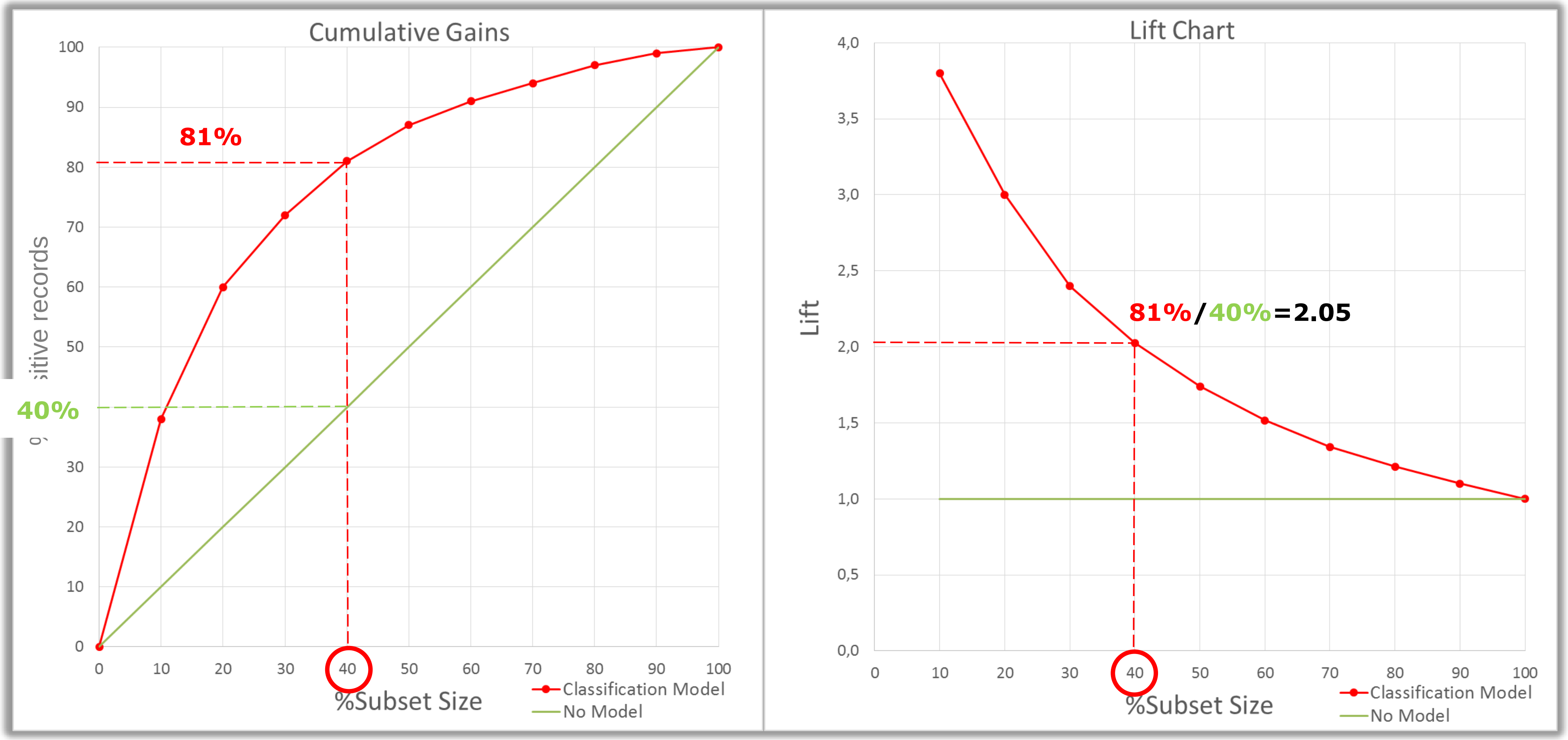
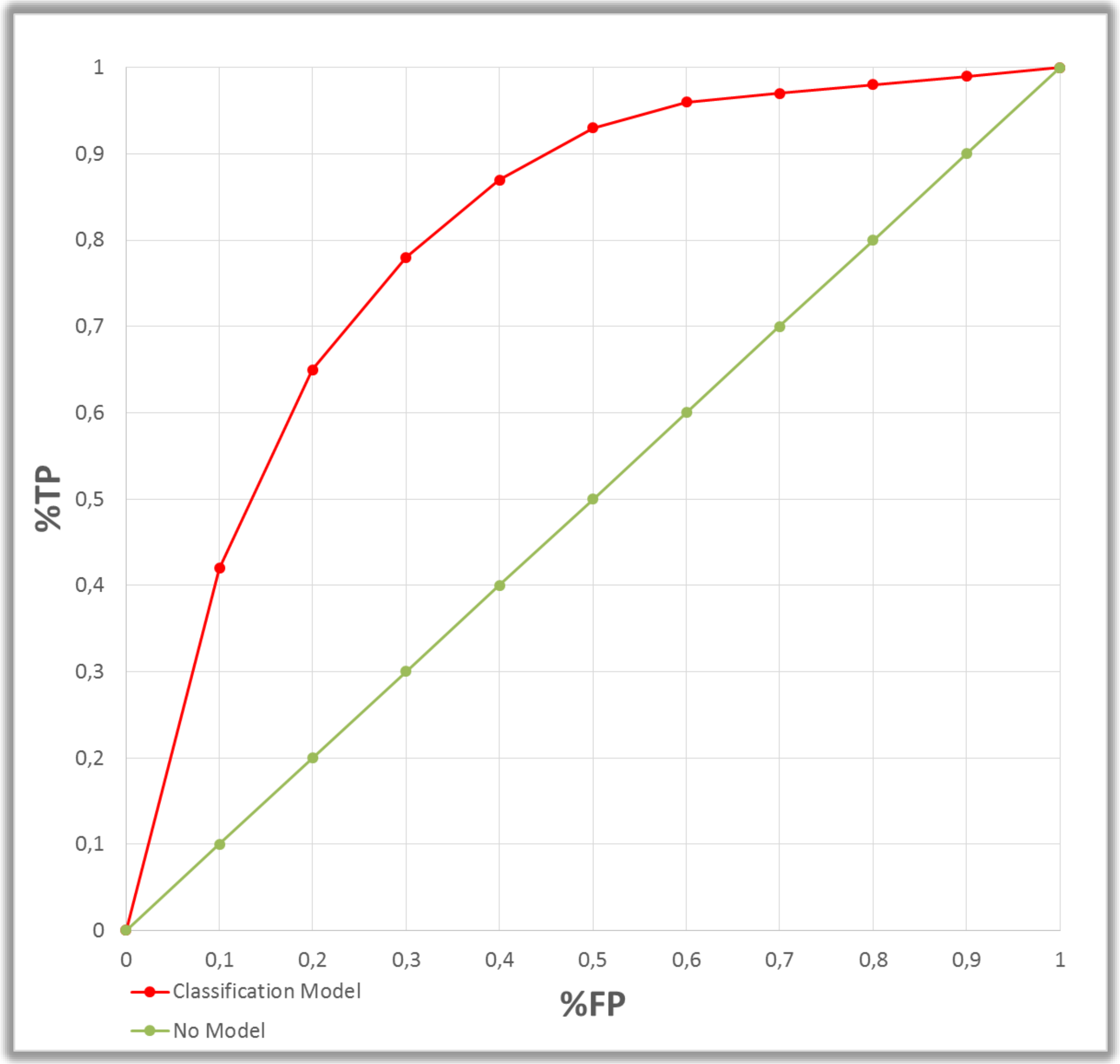


COUNTING THE COST: ROC CURVE

LIFT CHARTS are closely related to a graphical technique **FOR EVALUATING CLASSIFICATION MODELS** known as **RECEIVER OPERATING CHARACTERISTIC CURVE** (**ROC CURVE**).

ROC CURVE plots the **NUMBER OF POSITIVE RECORDS INCLUDED IN A SELECTED SUBSET OF RECORDS** on the vertical axis, **EXPRESSED AS PERCENTAGE OF THE TOTAL NUMBER OF POSITIVE RECORDS (%TP, TPR)**, **AGAINST THE NUMBER OF NEGATIVE RECORDS INCLUDED IN THE SUBSET**, **EXPRESSED AS A PERCENTAGE OF THE TOTAL NUMBER OF NEGATIVE RECORDS (%FP, FPR)**, on the horizontal axis.



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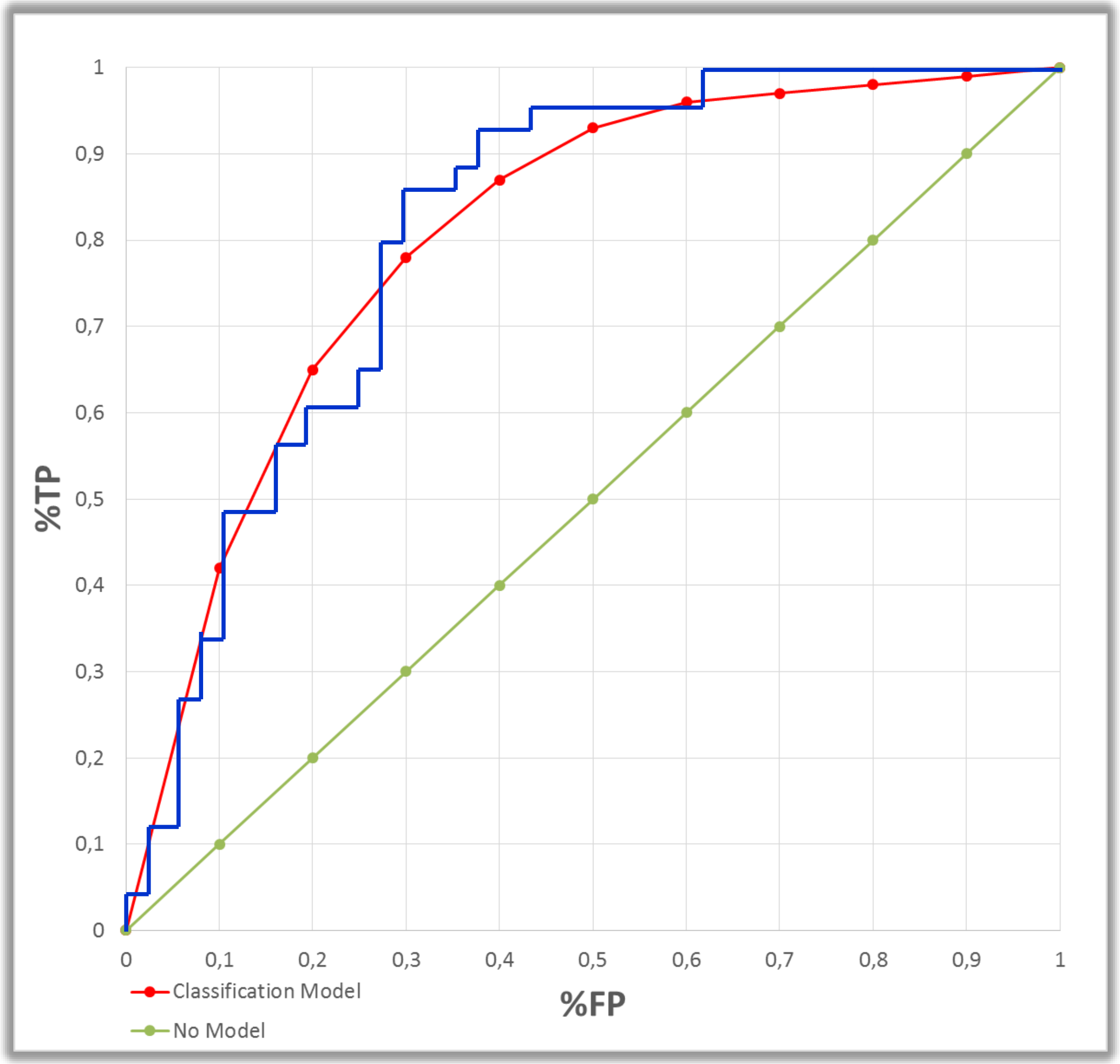
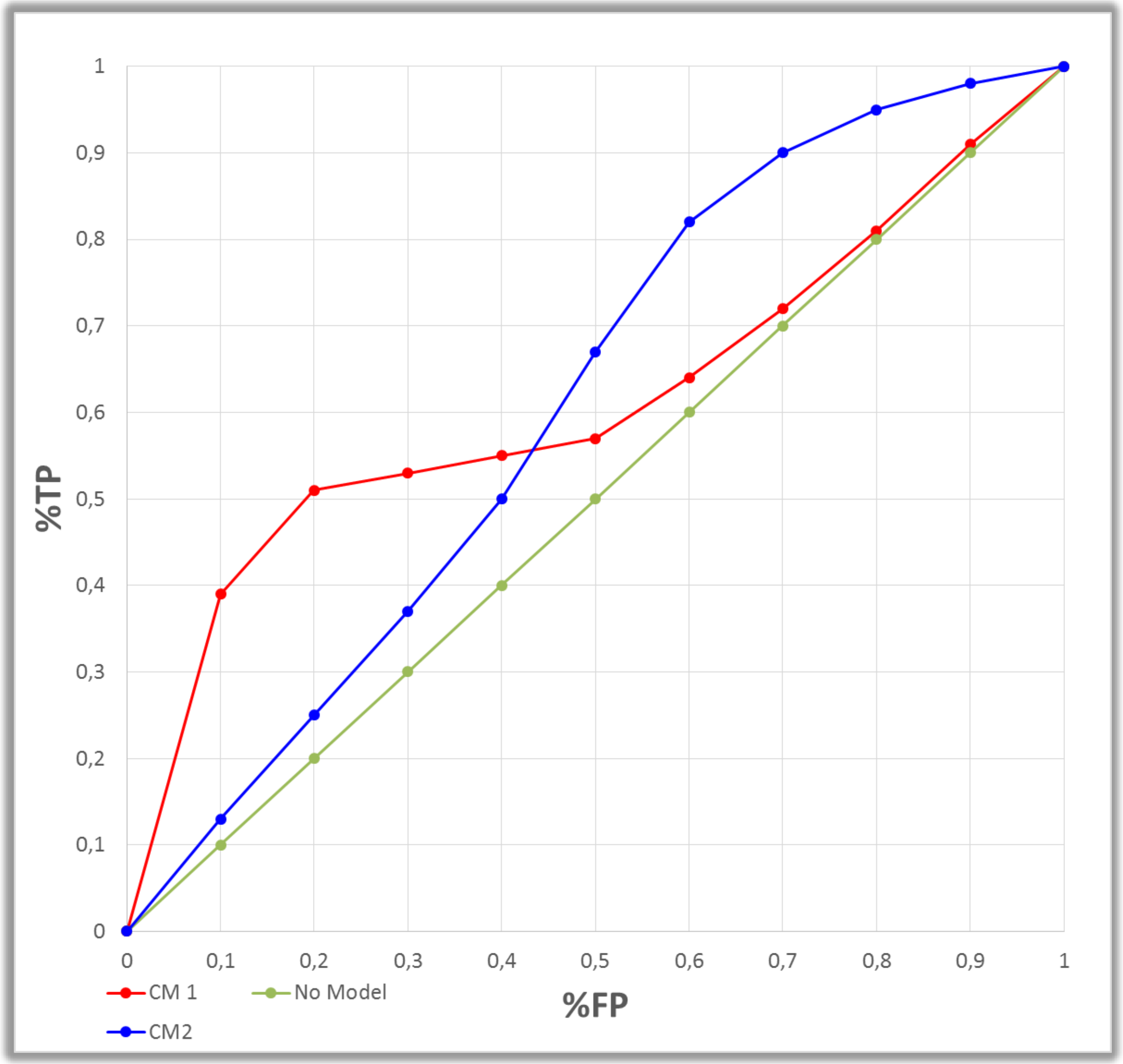
The **JAGGED ROC CURVE IN BLUE DEPENDS ON** the details of the **PARTICULAR SUBSET OF DATA**. This **DEPENDENCE** can be **REDUCED BY** applying **CROSS-VALIDATION**.

The **ROC CURVE DEPICTS THE PERFORMANCE OF A CLASSIFIER WITHOUT REGARD TO CLASS DISTRIBUTION OR ERROR COSTS.**



COUNTING THE COST: ROC CURVE

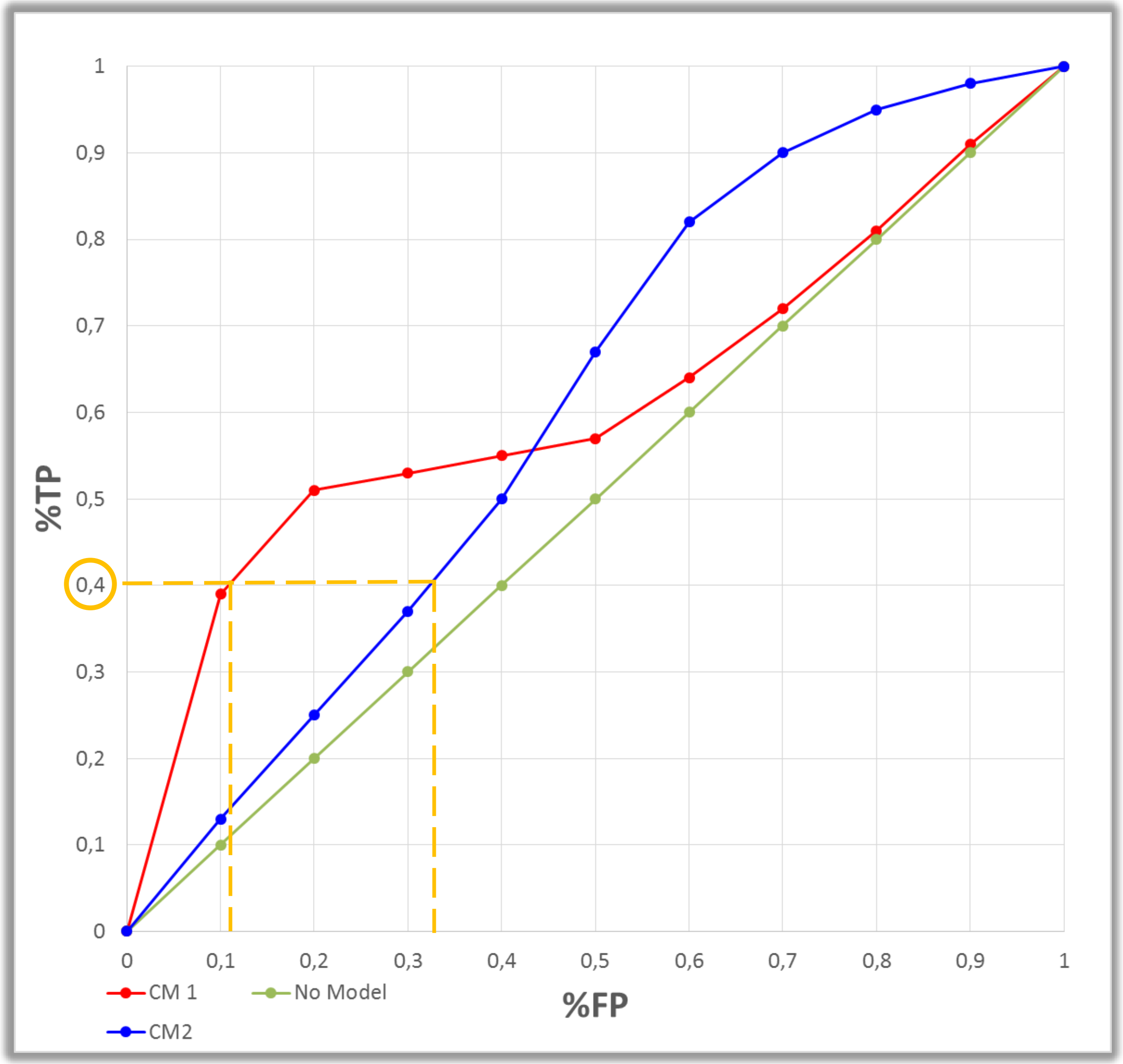
CM1 excels if a small, focused subset is sought; that is, if you are working towards the left-hand side of the graph.



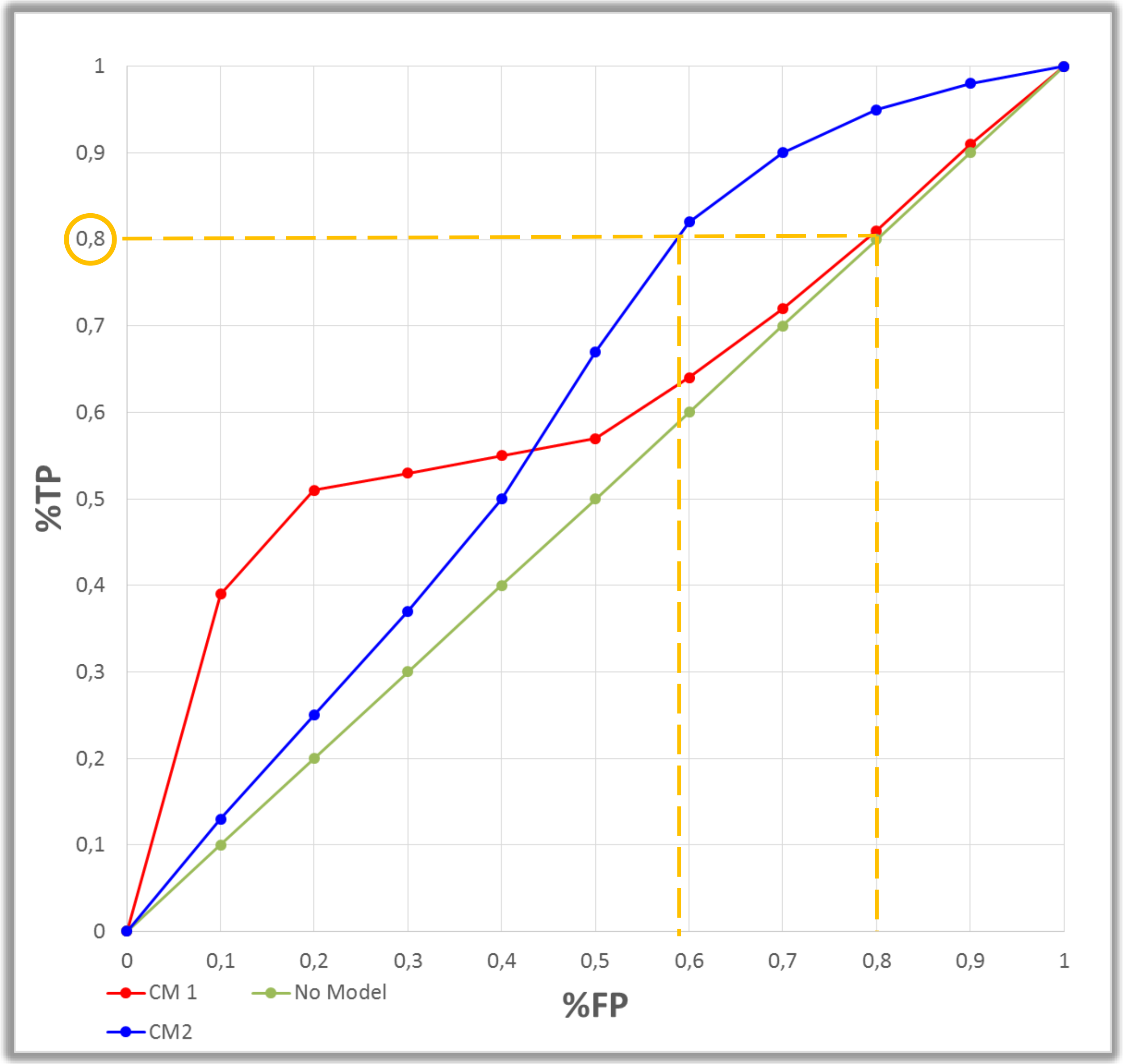
COUNTING THE COST: ROC CURVE

CM1 excels if a small, focused subset is sought; that is, if you are working towards the left-hand side of the graph.

Clearly, if you aim to cover just the 40% of the true positive records you should choose CM1, which gives a false positive rate of around 10%, rather than CM2, which gives over 30% false positive records.



CM2 excels if you are planning a large subset: if you are covering 80% of the true positive records, CM2 will give false positive rate around 60% as compared with the CM1's 80%.



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